MIL-P-60052 (MU)
28 August 1964

# MILITARY SPECIFICATION

## PRIMER, PERCUSSION, M1A2
## METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metals parts for one type of percussion primer designated as Primer, Percussion, M1A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenace of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235-- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1390

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DRAWINGS

ORDNANCE CORPS

8838079 - Head
8838089 - Body
8838012 - Cup, Battery
8838015 - Plug, Firing

PUBLICATIONS

ORDNANCE CORPS

IEL-7298952 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwgs) 8838079, 8838089, 8838012 and 8838015 as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws as, evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.4).

3.4 Workmanship.-All parts shall be dry and free of chips, cracks, dust, grease, rust, burrs and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier's responsibilities will be as specified in the contract or purchase



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order. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.6. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.
b. Number and revision date of specification.
c. Characteristic code number.
d. Lot or serial number.
e. Number of defectives found.
f. Assignable cause or analysis of occurrence.
g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of compliance on all material purchased



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by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantited by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of twenty-five assemblies and 25 each of every individual component and every sub-assembly which has been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All primer bodies and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in



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manufacturing process, material used or supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawing.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Product shall be submitted for inspection in accordance with Standard MIL-STD-105D (or Standard MIL-STD-1235, when applicable).

4.3.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard



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MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each minor defect and an AQL of 0.40 percent for each major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, battery (see dwg. B-8838012).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: None defined | | | |
| Major: | AQL 0.65 percent | | |
| 101. | Outside diameter, min. | Gage | 01001 |
| 102. | Thickness through bottom, maximum (max). | Gage | 01002 |
| 103. | Length. | Gage | 01003 |
| Minor: | AQL 2.50 percent | | |
| 201. | Diameter of hole. | Gage | 01004 |
| 202. | Thickness through bottom, min. | Gage | 01005 |
| 203. | Inside diameter. | Gage | 01006 |
| 204. | Outside diameter, max. | Gage | 01007 |
| 205. | Flat missing from surface at open end. | Visual | 01008 |
| 206. | Evidence of poor workmanship (see 3.4). | Visual | 01009 |

4.3.2.2 Plug, firing (see dwg. C-8838015).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |
| Major: | AQL 0.65 percent | | |
| 101. | Large diameter, min. | Gage | 02001 |
| 102. | Length and profile of point. | Gage | 02002 |
| 103. | Length of pilot, min. | Gage | 02003 |
| 104. | Total length. | Gage | 02004 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 2.50 percent | | |
| 201. | Large diameter, max | Gage | 02005 |
| 202. | Length of pilot, max | Gage | 02006 |
| 203. | Diameter of pilot | Gage | 02007 |
| 204. | Finish improper | Visual | 02008 |
| 205. | Evidence of poor workmanship (see 3.4) | Visual | 02009 |

4.3.2.3 Head (see dwg. 8858079).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 1.00 percent | | |
| 101. | Pitch diameter of body thread | Gage | 03001 |
| 102. | Major diameter of body thread, min | Gage | 03002 |
| 103. | Diameter of body | Gage | 03003 |
| 104. | Diameter of battery cup cavity | Gage | 03004 |
| 105. | Diameter of percussion element cavity | Gage | 03005 |
| 106. | Depth of percussion element cavity | Gage | 03006 |
| 107. | Thickness of flange, max | Gage | 03007 |
| 108. | Crack or split | Visual | 03008 |
| 109. | Flash hole missing or obstructed | Visual | 03009 |
| Minor: | AQL 6.50 percent | | |
| 201. | Major diameter of body thread, max | Gage | 03010 |
| 202. | Diameter of flange | Gage | 03011 |
| 203. | Depth of battery cup cavity | Gage | 03012 |
| 204. | Concentricity between percussion element cavity and flash hole | Gage | 03013 |
| 205. | Concentricity between body diameter and percussion element | Gage | 03014 |
| 206. | Concentricity between body diameter and flange | Gage | 03015 |



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| | | | |
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| 207. | Diameter of chamfer at mouth of percussion element cavity | Gage | 03016 |
| 208. | Diameter of flash hole | Gage | 03017 |
| 209. | Thickness of flange, min | Gage | 03018 |
| 210. | Length to shoulder above body thread | Gage | 03019 |
| 211. | Length of thread undercut, max | Gage | 03020 |
| 212. | Total length | Gage | 03021 |
| 213. | Radii missing | Visual | 03022 |
| 214. | Finish improper | Visual | 03023 |
| 215. | Evidence of poor workmanship (see 3.4) | Visual | 03024 |

4.3.2.4 Body (see dwg. 8838089).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Flash hole missing | Gage | 04001* |
| 2. | Crack or split | Visual | 04002* |
| Major: | AQL 1.00 percent | | |
| 101. | Pitch diameter of thread | Gage | 04003 |
| 102. | Length of thread, min | Gage | 04004 |
| 103. | Minor diameter of thread, max | Gage | 04005 |
| 104. | Wall thickness, min | Gage | 04006 |
| 105. | Thread stripped or otherwise damaged | Visual | 04007 |
| Minor: | AQL 4.00 percent | | |
| 201. | Length of thread, max | Gage | 04008 |
| 202. | Concentricity of inside diameter with pitch diameter of thread | Gage | 04009 |
| 203. | Thickness at closed end | Gage | 04010 |
| 204. | Location of first row of holes | Gage | 04011 |
| 205. | Inside diameter, min | Gage | 04012 |
| 206. | Depth of counterbore, max | Gage | 04013 |
| 207. | Diameter of counterbore | Gage | 04014 |



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|---|---|---|---|
| 208. | Total length, max | Gage | 04015 |
| 209. | Plug gage fails to enter body to required depth | Gage | 04016 |
| 210. | Body fails to pass through required ring gage | Gage | 04017 |
| 211. | Finish improper | Visual | 04018 |
| 212. | Hole in drawn end | Visual | 04019 |
| 213. | Evidence of poor workmanship (see 3.4) | Visual | 04020 |

4.3.3 Testing.-The test equipment shall be in accordance with 4.3.4.

4.3.3.1 Hydrostatic test (see 3.3) Critical defect - Code No. 04021.-This test shall be conducted 100 percent. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom. The test shall be performed as specified in 4.4.1.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Number IEL 7298952 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the



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equipment, and verification of it's accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Hydrostatic test.-Prior to machining, the tubing for primer fabrication shall be assembled into the test equipment specified in 4.3.4 and shall be tested for not less than five seconds at not less than the hydrostatic pressure calculated from the following formula:

$$P = \frac{2\ St}{D} \times .9$$

where:

P = test pressure in pounds per square inch (psi.).
S = minimum specified yield strength, psi.
T = specified manimum wall thickness in inches.
D = specified maximum outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data cards shall be prepared for each lot in accordance with information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection



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equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection



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equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above.



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6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Hydrostatic test.-If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.5 Submission of initial production samples.-Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly.

6.6 Critical defect report

6.6.1 Submission of critical defect reports.-For U. S. Army Munitions Command procurement, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUAP-RO; Picatinny Arsenal, ATTN: SMUPA-ND.

6.6.2 Reporting to other services.-For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the contracting officer.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A410



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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